

Vineet Agarwal

240-353-9811 — vineetagarwal540@gmail.com — vineetagarwal54 — Github — Portfolio

Summary — Software engineering graduate student with 2+ years shipping production iOS and Android apps in React Native and TypeScript, owning features from component architecture and state management through App Store and Play Store release. Comfortable across the stack, from diagnosing mobile performance regressions to designing Node.js and serverless AWS backends where cost, latency, and failure modes matter. Recent work spans real-time systems and on-device vision AI, building consumer products in which AI is one component within a larger experience.

Skills

Languages JavaScript, TypeScript, C++ , Python, SQL

Mobile React Native, Redux Toolkit, Component Architecture, Performance Profiling, Cross-Platform (iOS and Android), App Store and Play Store Release

On-Device AI on-device vision (Qwen3-VL), llama.rn, Whisper, RAG, OpenAI APIs

Real-Time & Data WebRTC, WebSocket, Socket.IO, MongoDB, Redis, Firebase (Push Notifications), Webhooks

Backend & Cloud Node.js, Express, REST APIs, AWS Serverless (Lambda, API Gateway, DynamoDB, SQS), PostgreSQL, Docker, CI/CD

Core Concepts Git, Agile Development, System Design, Cross-Platform Architecture, Performance Optimization

Experience

ServBeyond Solutions

Enterprise AI & Platform Intern

Jun 2026 – Present

Rockville, MD

- Shipped a company-wide GenAI assistant used daily by 20+ staff, wiring REST and SOAP integrations across Salesforce, ServiceNow, and AWS into a normalized data layer with a React front end.

Svipes

React Native Developer

Jul 2024 – Dec 2024

- Cut video playback stutter 70% and load time by 1.3 seconds through async loading and client-side caching, smoothing playback across iOS and Android.
- Shipped 50+ cross-platform React Native screens with Redux Toolkit, delivering modular UI systems that improved release consistency across both platforms.

Xelpmoc Design and Tech Limited

Software Developer

Nov 2022 – Apr 2024

- Built and shipped a production iOS and Android credentials wallet app; implemented 15+ blockchain webhooks enabling real-time verification workflows.
- Integrated Firebase and Twilio for push notifications and secure user communication, improving real-time engagement across platforms.
- Led mobile release cycles using Xcode and Android Studio; resolved critical UI defects and deployed stable updates to the App Store and Play Store.
- Developed twin React Native apps for an old-age home network, improving operational coordination by more than 60% through real-time task sync.

Education

University of Maryland, College Park

Master of Engineering , Computer Software Engineering

Jan 2025 - Dec 2026

- Minor in Cloud Engineering
- Relevant Coursework: Software Design, Cloud Computing, Virtualization, Cloud Security

Projects

Locra: On-Device Multimodal AI, Android (Github)

Jun 2025 – Present

- Built an Android-first multimodal AI assistant in React Native, TypeScript, and Expo that runs a quantized Qwen3-VL-2B vision-language model fully on device via llama.rn, with no internet needed after model download.
- Added on-device speech to text with Whisper (whisper.rn) and live capture via Vision Camera, and enforced a hard offline guarantee through a network gate and a Kotlin native module that verifies model integrity before inference.

Serverless Video Analytics Platform, AWS (Github)

Sept 2023 – Dec 2023

- Designed a serverless pipeline on AWS Lambda, API Gateway, DynamoDB, and SQS with CloudFormation and GitHub Actions, using dedicated upload, processing, and analytics functions with auto-scaling behind CloudFront to absorb bursty load at low idle cost.